Java Programming Project Presentation

KG COLLEGE OF ARTS AND SCIENCE
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KGiSL Campus, Saravanampatti, Coimbatore – 641 035

- Project Title: Semester GPA & CGPA Calculation System
- Course: java Programming Mini Project
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- Department: Computer Science

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Introduction

- GPA (Grade Point Average) measures performance in a semester.
- CGPA (Cumulative GPA) measures overall academic performance.
- This system automates calculation.

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Objective

- Calculate Semester GPA
- Calculate CGPA
- Reduce manual errors
- Provide quick results

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Problem Statement

Students often struggle with calculating and tracking their semester GPA and overall CGPA. Manual methods are time-consuming and prone to errors, making it difficult to monitor academic progress accurately.
There is a need for a simple system that:

- Takes grades and credit hours as input.
- Calculates GPA for each semester and CGPA across semesters.
- Provides clear, error-free results in an easy-to-use format.

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System Requirements

- Hardware Requirements:
- • Processor: Intel i3 or higher
- • RAM: 4 GB minimum
- • Storage: 500 MB free space

- Software Requirements:
- • OS: Windows / Linux
- • Compiler: GCC / Turbo C / Code::Blocks / Dev C++

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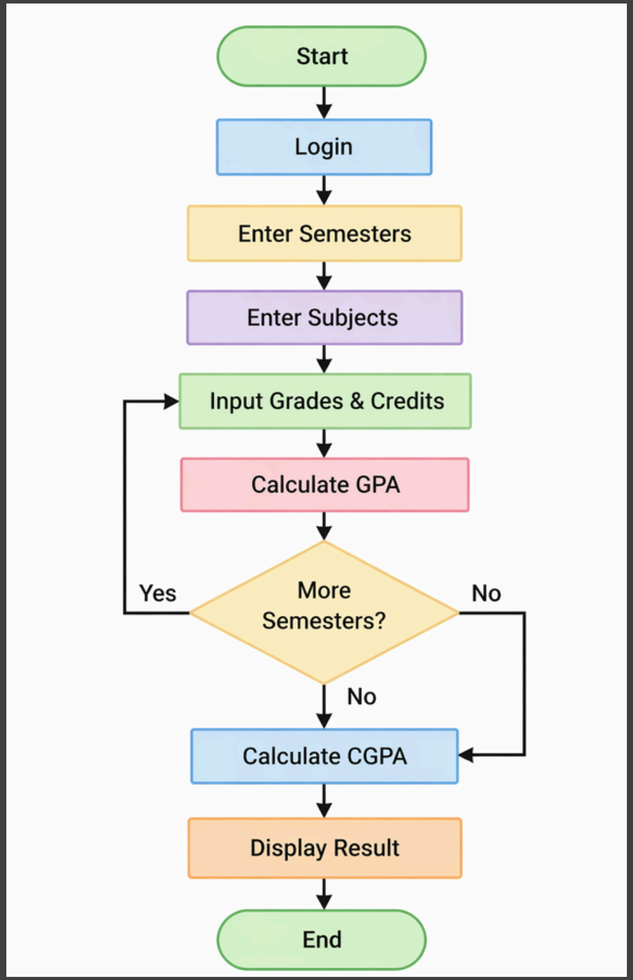
System Design



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Module Description

1. Input Module – Collects grades and credit hours.
2. Calculation Module – Computes semester GPA and cumulative CGPA.
3. Database Module – Stores and manages student records.
4. Report Module – Generates summaries and performance charts.
5. Interface Module – Provides a simple, user-friendly front end.

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Algorithm / Flowchart

1. Start
2. Input grades and credit hours
3. Convert grades → grade points
4. Calculate GPA = (Σ grade points × credits) ÷ (Σ credits)
5. Calculate CGPA = (Σ semester GPA × credits) ÷ (Σ total credits)
6. Display GPA and CGPA
7. End

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Implementation

Input – Students enter grades and credit hours.

1. Calculation – System computes GPA per semester and overall CGPA.
2. Database – Records are stored and updated securely.
3. Reports – Summaries and charts show performance.
4. Interface – Simple front end for easy use.

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Output Screens

1. Login Screen – Allows students and administrators to securely log in using username and password.
2. Input Screen – Enables users to enter subject grades and credit hours.
3. Calculation Screen – Computes and displays GPA and CGPA based on entered data.
4. Database Screen – Stores and shows student academic records.
5. Report Screen – Presents performance summaries using tables or charts.
6. Dashboard Screen – Provides a clear overview of academic progress and performance.
7. About Screen – Displays information about the project, its purpose, and technologies used.

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Results & Discussion

Summary of Outputs

- Calculates GPA & CGPA correctly
- Displays performance clearly

Comparison with Expected Results

- Matches manual calculations
- Results are correct

- Accuracy & Performance
- High accuracy
- Fast and efficient

Challenges & Solutions

- Input errors → fixed using validation
- Accuracy issues → used correct formulas
- Multiple semesters → handled using loops

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Conclusion & Future Enhancement

Conclusion

- Project successfully achieved its objectives
- Improved understanding of Java concepts and system design

Future Enhancements

- Add GUI support
- Integrate database
- Extend features for scalability

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References / Acknowledgement

- Java Programming – Herbert Schildt Java: The Complete Reference
- Oracle Java Documentation
- GeeksforGeeks – Java Concepts & Examples
- W3Schools – Basic Java Syntax
- Lecture notes and class materials

Thank You

Discussion

- What is GPA?
- What is CGPA?
- What is the difference between GPA and CGPA?